A lack of robust cross-domain structural priming

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Domains like language, mathematics, and music rely on hierarchical structural relationships between elements to establish the overall meaning of a sentence, expression, or melody. Processing information in these domains likely requires mental representation of abstract structure. Evidence for abstract structural representations in language comes from decades of structural priming research showing exposure to a structure facilitates future processing/production of that structure (e.g., Pickering & Ferreira, 2008). More recently, research has been devoted to determining whether structural representations are so abstract that representations in one domain (e.g., mathematics) influence structural processing in another domain (e.g., language). For example, Scheepers et al. (2011) found that solving a mathematical expression with a low (or high) attachment mathematical structure increased the likelihood that participants would complete a sentence with a low (or high) attachment syntactic structure. This finding (see also Scheepers et al., 2019; Van de Cavey & Hartsuiker, 2016) supports the existence of highly abstract, domain-general structural representations, relevant to theoretical frameworks in several cognitive domains. However, there are relatively few (< 10) published papers on this effect, and the original study has yet to be directly replicated. Thus, the current study sought to replicate the cross-domain priming results from Scheepers et al. in both online and in-person contexts, in speakers of American English.

Experiment 1 (E1) was administered online via Qualtrics, using the original design and stimuli from Scheepers et al. (2011). Participants (N = 119) solved 24 prime math expressions, 8 in each condition: Baseline (no grouping/solved left to right: 7 + 24 = ?), High Attachment (middle elements combined first: 7 + (28 − 4) × 2 = ?), or Low Attachment (final elements combined first: 7 + 28 – (4 × 2) = ?). Item condition was counterbalanced across 6 lists. Each prime expression was presented alone on the screen until the participant typed an answer. Next, a target sentence stem (e.g., *The reporter waited for the musicians of the pop star who*) appeared on the screen until the participant typed a sentence continuation. 2-3 filler sentences/expressions intervened between each prime-target pair. Experiment 2 (E2) was the same as E1, except it was administered in person using paper booklets, with one expression or sentence per page, completed sequentially (as in Scheepers et al., 2011). Participants (N = 154) completed the study in a quiet room with < 30 other participants. Calculators were not permitted in either experiment and all participants were recruited from university Calculus classes.

Target responses referring to the first noun after the verb (e.g., *who… were tuning their instruments*) were coded as high attachment structures, and those referring to the second noun after the verb (e.g., *who… was always late*) were coded as low attachment structures for our binary dependent variable (see Figure 1). Targets left blank, following incorrectly solved primes, or that had non- high or low attachment structure, were excluded (~20% of E1 trials; ~19% of E2 trials). To test whether the structure of the math prime influenced the structure of the target sentence, logistic mixed-effects models with target structure as the outcome variable and treatment-coded prime Condition as the fixed effect (participants and items included as random intercepts) were run on the data from each experiment, and on the combined data from both experiments. No significant effects were obtained in the models for E1, E2, or in the combined model (See Tables 1-3). A Bayesian hierarchical model and Bayes Factor estimate (BF = .23) for the combined data with mildly informative priors estimates the null model (without the Condition variable) was 4.3 times more likely than the alternative.

The current study failed to replicate Scheepers et al.'s (2011) cross domain priming effects using the same materials and a combined math student sample 2-3 times larger than originally tested. These results suggest that the robustness of cross-domain priming effects requires additional study, particularly as they pertain to development of cognitive and computational models of structural representation and processing (e.g., Johns et al., 2020).

Figure 1. Proportion of low attachment (LA) target completions, broken down by prime condition and experiment (Experiment 1 on the right and Experiment 2 on the left)

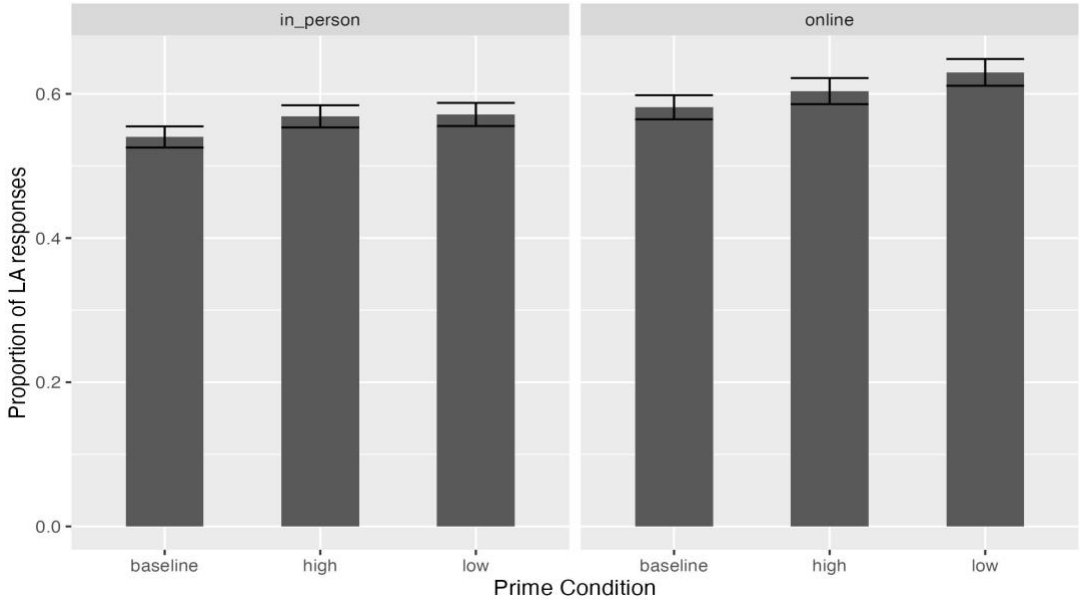

Table 1. Model output from Experiment 1 (online, *N* = 119) in log-odds

	Estimate	SE	z-value	p-value
Intercept (Baseline)	0.373	0.21	1.75	0.08
High Attachment Prime	0.092	0.12	0.80	0.43
Low Attachment Prime	0.151	0.12	1.28	0.20

Table 2. Model output from Experiment 2 (in person, *N* = 154) in log-odds

	Estimate	SE	z-value	p-value
Intercept (Baseline)	0.208	0.20	1.04	0.30
High Attachment Prime	0.077	0.10	0.80	0.43
Low Attachment Prime	0.096	0.10	0.98	0.33

Table 3. Model output from Experiments 1-2 combined (*N* = 273) in log-odds

	Estimate	SE	z-value	p-value
Intercept	0.203	0.20	1.01	0.31
High Attachment Prime	0.068	0.10	0.71	0.48
Low Attachment Prime	0.086	0.10	0.88	0.38
Experiment	0.171	0.13	1.36	0.18
High Attachment Prime X Experiment	0.012	0.15	0.08	0.93
Low Attachment Prime X Experiment	0.078	0.15	0.51	0.61

Table 4. Bayesian model estimating effect of Condition, combined data (*N* = 273) in log-odds

	Estimate	Est. Error	Lower Bound 95% Credible Interval	Upper Bound 95% Credible Interval	R-hat
Intercept (Baseline)	0.27	0.13	0.03	0.53	1.00
High Attachment Prime	0.06	0.07	-0.07	0.20	1.00
Low Attachment Prime	0.11	0.07	-0.03	0.25	1.00